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# Parallel electromembrane extraction of peptides with monoterpene and medium-length fatty acid deep eutectic solvents

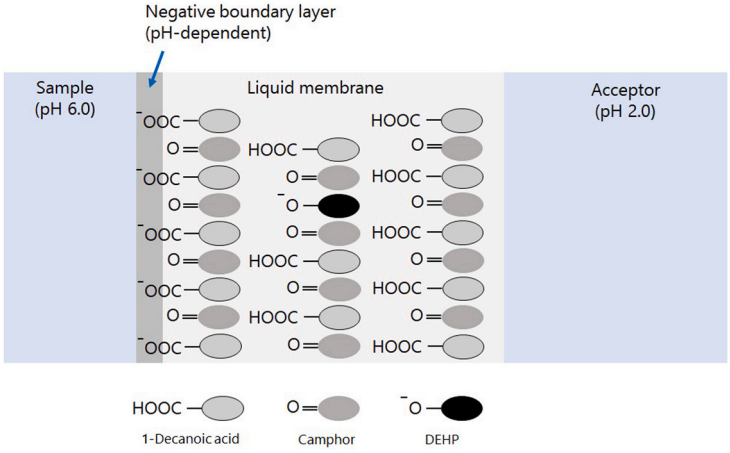
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## HIGHLIGHTS

- Camphor and 1-decanoic acid containing 2% di (2-ethylhexyl) phosphate (DEHP) was used as liquid membrane for electromembrane extraction of peptides.
- With this liquid membrane, the current across the liquid membrane was low and the system was highly stable during extraction from human plasma.
- Recoveries exceeding 40% was obtained for nine of 13 model peptides included in our training set of analytes.

## GRAPHICAL ABSTRACT




## ARTICLE INFO

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## ABSTRACT

*Background:* Electromembrane extraction (EME) involves the process of mass transfer of charged analytes from an aqueous sample through an organic liquid membrane into an aqueous acceptor medium under the influence of an electrical field. Successful solvation of the analyte within the liquid membrane is of paramount importance and involves molecular interactions with the liquid membrane. In this comprehensive investigation, parallel EME was examined using a training set of 13 model peptides employing deep eutectic solvents as the liquid membrane. These deep eutectic solvents were formulated by mixing specific monoterpenes (thymol, menthol, camphor) with medium-chain fatty acids (1-octanoic acid and 1-decanoic acid).
*Results:* From an array of different liquid membrane compositions explored, it was revealed that the combination of camphor and 1-decanoic acid (in a 1:1 w/w ratio) with 2% di (2-ethylhexyl) phosphate (DEHP) delivered the most efficient extraction system. The solvation of the model peptides within this liquid membrane predominantly relied on ionic interactions between protonated basic functionalities and DEHP, along with hydrogen bond

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interactions between the deprotonated acid functionalities (hydrogen bond acceptor) and 1-decanoic acid (hydrogen bond donor). Selectivity was modulated by the pH of the sample and acceptor solutions, with a direct correlation to the polarity and net charge of the model peptides. The ionization of 1-decanoic acid in the interfacial region between the sample and liquid membrane emerged as an important factor influencing the selectivity.

*Significance and novelty:* Although parallel EME of peptides has been reported previously, the current liquid membrane provides an extraction system with sufficient stability for the first time. Selective extraction of peptides through EME holds substantial promise within the realm of next-generation environmentally-friendly sample preparation methodologies. The findings presented in this paper contribute significantly to our fundamental understanding of these processes, and may serve as an important reference for the development of future methods in this field.

## 1. Introduction

Parallel electromembrane extraction (EME), a recently commercialized technique, represents an innovative and environmentally friendly approach for extracting bases, acids, and ionic substances [1,2]. EME functions as a three-phase system, wherein charged compounds migrate from the sample across an organic liquid membrane and ultimately enter an aqueous acceptor solution. This migration is driven by an applied electric potential across the liquid membrane. Notably, the direction of the electric field can be selected to extract either positively or negatively charged analytes. The selectivity of the extraction process is finely tuned through pH adjustments in both the sample and acceptor, as well as by the choice of organic solvent for the liquid membrane. EME, as a sample preparation method, promises rapid extraction with high selectivity and potential for pre-concentration [3–5]. Furthermore, EME requires minimal consumption of organic solvents, chemicals, and reagents, aligning with the green principles for sample preparation [6]. Similarly to other liquid membrane-based approaches [7,8], EME has undergone advancements in automation, further enhancing its reproducibility. Although automation of is still in its early stages of development, these EME systems offer the advantage of reduced manual intervention, minimizing the risk of errors and improving overall method robustness [9–11].

Given its versatility, EME has been applied in a wide array of fields, including the extraction of pharmaceuticals from biological fluids [12–14], environmental pollutants [15,16], and contaminants from food and beverage samples [17–19]. Peptide extraction via EME has been explored using two distinct approaches: traditional EME employing supported liquid membranes and gel-EME. In the latter approach, a gel membrane, typically composed of agarose, separates the sample and acceptor phases [20]. Gel-EME excels in its aqueous nature, enabling the extraction of polar substances with a high recovery. However, it is limited in terms of selectivity, and during extraction, the volumes of the sample and acceptor phases change due to electroendosmosis.

Alternatively, peptides have been extracted in traditional EME systems utilizing an organic solvent immobilized in a supporting membrane as the liquid membrane [21–29]. These EME systems offer high selectivity and are not susceptible to electroendosmosis. However, the challenge lies in the extraction of polar peptides across organic liquid membranes. Initial studies employed mixtures of 1-octanol and di (2-ethylhexyl) phosphate (DEHP) as the liquid membrane, with DEHP serving as a carrier to facilitate solvation of peptides in the liquid membrane by ionic interactions [21–25]. While efficient extractions were achieved from pure water samples, challenges arose when extracting from human plasma samples due to excessive current (>50 μA). Excessive current introduces instability into the extraction system, leading to pH changes in both the sample and acceptor phases as a result of electrolysis. Additionally, gas generation from electrolysis may elevate system pressure. Subsequent research aimed to mitigate this issue by mixing 1-octanol and DEHP with di-isobutyl ketone [26] or by using 1-nonanol mixed with 2-decanone and DEHP [27]. In addition, the EME of peptides without ionic carriers have been reported using tri (pentyl) phosphate as the liquid membrane [28]. While the shift towards lower polarity liquid membranes reduced the extraction current and improved the system robustness, it came at the expense of extraction efficiency.

Recently, EME of peptides were demonstrated for the first time using a deep eutectic solvent mixed with DEHP as the liquid membrane [30]. Due to its strong hydrogen bond acceptor and donor properties, several eutectic compositions exhibited superior efficiency for peptide extraction compared to the previously discussed liquid membranes. However, these eutectic solvents posed compatibility challenges with biological samples and led to very high extraction currents.

The present study reports on parallel EME of peptides employing a novel category of deep eutectic solvents derived from monoterpenes (thymol, menthol, camphor) and medium-chain fatty acids (1-octanoic acid and 1-decanoic acid). Unexpectedly, one of these eutectic compositions displayed remarkable efficiency from plasma, and provided a stable extraction system. The extraction performance of various eutectic compositions is discussed, to understand the fundamental aspects and requirements for EME of peptides. Furthermore, extraction with the most promising eutectic composition was optimized, and the performance and selectivity of this system are discussed in detail. We foresee that EME can be developed as a highly valuable tool for the extraction of peptides from biological fluids, including signature peptides originating from proteins subjected to enzymatic digestion. This fundamental study represents an important step in this direction of research.

## 2. Experimental

### 2.1. Chemicals and reagents

The peptides, including angiotensin I trifluoroacetate, angiotensin II acetate, angiotensin III, angiotensin IV, neurotensin acetate, endomorphin-1 trifluoroacetate, bradykinin acetate, (arg8)-vasopressin trifluoroacetate, oxytocin acetate, leu-enkephalin acetate, met-enkephalin acetate, diprotin A (Ile-Pro-Ile), and diprotin B (Val-Pro-Leu) were provided by Bachem (Bubendorf, Switzerland). Formic acid, phosphoric acid, di (2-ethylhexyl) phosphate, thymol, menthol, camphor, 1-octanoic acid, and 1-decanoic acid were supplied by Sigma-Aldrich (St. Louis, MO, USA). LC-MS grade acetonitrile and methanol were obtained from Sigma-Aldrich. Deionized water was obtained using a Milipak® (0.22 μm filter) Milli-Q water purification system (Milli-Q, Molsheim, France). The circular polypropylene membrane with a thickness of 110 μm and a diameter of 9 mm was obtained from Extraction Technologies Norway (Ski, Norway). Stock and working solutions of the peptides were prepared according to our previous study for daily use [28].

### 2.2. Preparation of deep eutectic solvents

DESs were prepared by weighing appropriate amounts, typically 1 g of each component into a 5 mL Eppendorf-tube. After weighing, the tube was heated in an oven at 65 °C for approximately 15 min to facilitate melting. After melting, the mixture was vortexed briefly.

### 2.3. Liquid chromatography-tandem mass spectrometry (LC-MS/MS)

LC-MS/MS analysis was conducted utilizing an Agilent 1290 Infinity II UHPLC system (Agilent Technologies, Santa Clara, CA, USA), equipped with a binary pump and an auto-sampler. To separate the analytes, an Eclipse C18 column (1.0 × 50 mm, 3.0 μm particles, Thermo Scientific) was employed and maintained at a temperature of 40 °C. A gradient elution method was employed, utilizing mobile phase A (95:5 v/v deionized water and acetonitrile containing 0.1% formic acid) and mobile phase B (5:95 v/v deionized water and acetonitrile containing 0.1% formic acid). The gradient began with 3% B at 0.00 min, increased linearly to 5% B from 0.00 to 0.50 min. It increased linearly to 30% B from 0.50 to 2.50 min, followed by a switch to 100% B from 2.51 to 3.00 min, and finally returned to 3% B from 3.00 to 3.01 min. The flow rate remained constant at 0.6 mL $min^{-1}$. The analytes were detected using an Agilent 6495 LC/TQ (Agilent Technologies) mass spectrometer with positive electrospray ionization at 3 kV. The system was utilized in the dynamic MRM mode, as reported previously [28].

### 2.4. Liquid chromatography – ultraviolet detection (LC-UV)

Camphor and decanoic acid were quantified using a Dionex UltiMate 3000 RS UHPLC system (Thermo Scientific, San Jose, CA, USA) comprising a pump, an auto-sampler, a temperature-controlled column compartment, and UV-detector. An Acquity UPLC® HSS T3 column (100 × 2.1 mm ID, 1.8 μm, Waters, Wexford, Ireland) maintained at 60 °C was used. The mobile phase consisted of A) 95:5 v/v deionized water and acetonitrile with 5 mM phosphoric acid and B) 5:95 v/v deionized water and acetonitrile with 5 mM phosphoric acid, using a flow of 0.5 mL $min^{-1}$ and gradient elution. The gradient was increased linearly from 20% to 100% B in 0.0–5.0 min, and held at 100% B from 5.0 to 6.0 min, before returned to 20% B in 0.1 min for a final 4 min of re-equilibration. Camphor and decanoic acid were detected at 280 and 210 nm, respectively.

#### 2.4.1. Parallel EME equipment

In this study, EME experiments were conducted using a vial-based prototype device (Fig. 1) provided by Extraction Technologies Norway (Ski, Norway). EME vials, made of a conducting polymer, served as electrodes and compartments to hold 300 μL of the sample and acceptor solutions. To connect the sample and acceptor vials, a circular porous polypropylene membrane (Extraction Technologies, Norway) was placed in a tailor-made union designed for the supporting membranes.

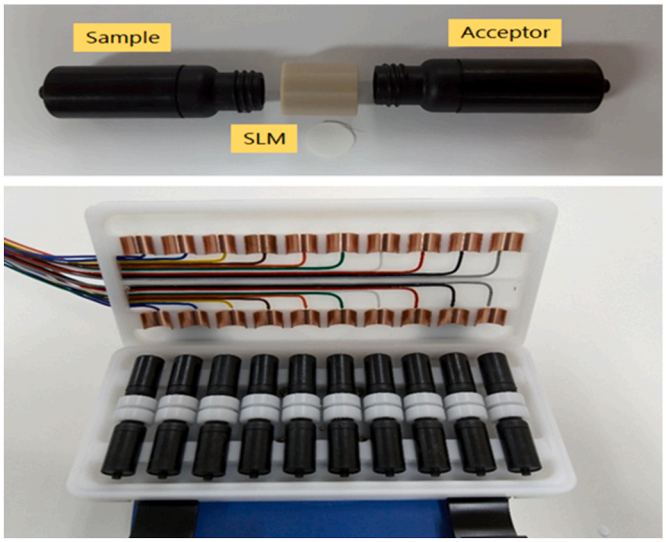



**Fig. 1.** Photo of equipment for electromembrane extraction.

In this way, the union served as a framework for assembling the entire extraction cell, connecting the sample vial, liquid membrane, and acceptor vial. The extraction cell, comprising all these components, was then positioned in a holder (ten-position holder). Thus, the conducting sample and acceptor vials encountered electrodes coupled to an external power supply. This configuration enabled simultaneous extraction of up to ten samples. To facilitate agitation during extraction, the ten-position holder was mounted on a shaker, ensuring the samples were adequately mixed. To apply voltage, a power supply Model ES 0300-0.45 from Delta Elektronika BV (Zierikzee, Netherlands) was used. Changes in the extraction current were monitored using a Fluke 287 multi-meter (Everett, WA, USA).

*2.4.1.1. Plasma samples.* Plasma from healthy blood donors were obtained from Oslo University Hospital (Oslo, Norway), and kept at −28 °C. Frozen plasma was quickly thawed at room temperature and then mixed with the model peptides to achieve a concentration of 2.0 μg/mL. Prior to extraction, plasma samples were diluted 1:1 (v/v) with 25 mM phosphate/citrate buffer to the desired pH.

### 2.5. Parallel EME procedure

First, 300 μL of plasma sample was pipetted into the conductive sample vial and 300 μL of acceptor was pipetted into the conductive acceptor vial. Next, the circular membrane was inserted into the tailor-made union. To impregnate the solid membrane, 9.0 μL of the liquid membrane was pipetted onto the acceptor side of the membrane. Finally, the vials were connected using the tailor-made union to establish the entire extraction cell. Up to ten extraction cells were placed in the ten-position holder. When the lid of the ten-position holder was fixed, the positive electrode (anode) was connected to the sample vials, whereas the negative electrode (cathode) was connected to the acceptor vials. The ten-position holder was then placed on the shaker. The extraction process was initiated by applying a voltage of 15 V and shaking the mixture at 900 rpm. The acceptors were collected after 45 min of extraction and analyzed by LC-MS/MS. The extraction current was measured to assess the system stability.

### 2.6. Calculations

For each analyte, recovery (R) was calculated using the following equation:

$$R = \frac{n_a}{n_d} \bullet 100\% = \frac{V_a\, C_{a,final}}{V_d C_{d,initial}}\, x\ 100\% \quad (1)$$

In this context, $n_a$ and $n_d$ represented the moles of analyte ultimately collected in the acceptor and initially present in the sample, respectively. $C_{a,final}$ and $C_{d,initial}$ denote the final and initial concentrations of the analyte in the acceptor and sample solutions, respectively. $V_a$ and $V_d$ refer to the volumes of the acceptor and sample, respectively.

The matrix effect (ME%) quantifies the impact of ion suppression during the electrospray ionization process, and was calculated using the following equation:

$$ME\ (\%) = \frac{peak\ area\ of\ post - extraction\ spiked\ matrix}{peak\ area\ of\ unextracted\ standard}\ x\ 100\% \quad (2)$$

## 3. Results and discussion

### 3.1. Theoretical considerations and design of experiments

Peptides are predominantly polar molecules, characterized by the presence of multiple positive and negative charges. Electromembrane extraction (EME) of such substances encounter significant challenges due to the limited solvation and partitioning behavior of peptides within the organic liquid membrane. Solvation is facilitated by molecular

interactions between peptides and the solvent comprising the liquid membrane. Ionic, hydrogen bond, and cation-π interactions are recognized as primary contributors. In the current EME literature, peptide extraction typically occurs under moderately acidic conditions in the sample and acceptor ($2.0 < pH < 4.0$), with an anionic carrier, such as di (2-ethylhexyl) phosphate (DEHP), dissolved in the liquid membrane. Under these conditions, the base functionalities of peptides are fully protonated, and ionic interactions with the carrier contribute to solvation. Moreover, the positive charges serve as hydrogen bond donors and interact by hydrogen bond interactions with the liquid membrane if it has sufficient hydrogen bond basicity. The positive charges also interact with the membrane solvent by cation-π interactions, provided that the latter is aromatic. The acidic functionalities of peptides are partially or fully charged under moderately acidic conditions and provide hydrogen bond basicity. For proper solvation, the liquid membrane should also exhibit a certain level of hydrogen bond acidity.

Hence, it is imperative to consider properties such as hydrogen bond basicity, hydrogen bond acidity, and π-donor characteristics when designing liquid membranes for EME of peptides. Additionally, the hydrophobicity of the liquid membrane plays a critical role in this process. Liquid membranes with low hydrophobicity (log $P < 2.5$) are susceptible to leakage. Moreover, such liquid membranes often exhibit high electrical currents, due to their permeability to a wide range of ionic species present in the sample. High current (>50 μA per sample) causes excessive electrolysis within the sample and acceptor, leading to undesirable consequences such as pH drift and increased pressure. Conversely, very hydrophobic liquid membranes (log $P > 5.0$), which are effective in preventing leakage, are impermeable to peptides, limiting their utility in peptide extraction processes.

In this study, hydrophobic deep eutectic solvents were investigated by employing monoterpenes such as thymol, menthol, and camphor as hydrogen bond acceptors, in combination with medium-chain fatty acids, specifically 1-octanoic, 1-nonanoic, and 1-decanoic acids, as hydrogen bond donors (Table 1). The preparation of deep eutectic solvents followed the experimental procedure outlined in the Experimental section, and their suitability as liquid membranes for the extraction of 13 distinct model peptides was thoroughly examined. Relevant physicochemical properties of the model peptides within this training set of substances are summarized in Table S1 (supporting information).

Parallel EME was conducted, using pure buffer solutions and plasma samples spiked with the model peptides. Continuous monitoring of the electrical current during EME allowed the construction of current profiles. These profiles served as a crucial tool for assessing the stability of the extraction system. Stability was deemed satisfactory when the current per sample was below 50 μA throughout the entire extraction process [30].

The acceptor phases were subjected to LC-MS/MS analysis after extraction, enabling the calculation of extraction recoveries. Recoveries exceeding 40% were considered acceptable, while those exceeding 85% were indicative of exhaustive extraction [30]. In the case of buffer solutions, post-extraction, the samples underwent LC-MS/MS analysis to establish a mass balance for each model peptide.

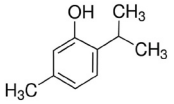
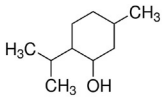
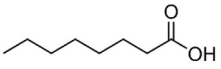
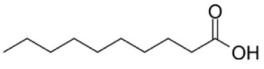
### 3.2. Extraction with thymol as hydrogen bond acceptor

In the initial set of experiments, extractions were conducted using a deep eutectic solvent derived from thymol and 1-octanoic acid, employing spiked buffer solutions as the sample. The sample and acceptor phases were adjusted to pH 3.0 and 2.0, respectively. The EME was conducted at 30 V for 30 min. Surprisingly, thymol-octanoic acid solvent exhibited exceptional extraction performance when applied to buffer solutions. Extraction recoveries were notably high, with 11 peptides showing recoveries exceeding 70% (Table S2 in supporting information).

Kamlet-Taft values for the thymol-octanoic acid solvent were measured to be $\alpha = 0.9$, $\beta = 0.1$, and $\pi^* = 0.9$, using the same method as in a previous study [30]. Notably, the high α-value of the membrane solvent for hydrogen bond acidity indicated a significant contribution of hydrogen bond interactions with negative charges of the peptides. Conversely, the membrane solvent exhibited low hydrogen bond basicity (β), suggesting that hydrogen bond interactions with positive charges of the peptides were of minor importance. The interactions between the membrane solvent and the protonated base functionalities were likely driven by cation-π interactions, thus elucidating the remarkable extraction performance of thymol-octanoic acid due to its aromatic character and strong hydrogen bond donor (HBD) properties.

Subsequently, we examined the suitability of thymol and 1-octanoic acid with spiked plasma samples. Compared to previous experiments with pure buffer solutions, there was a significant increase in the extraction current, and the stability of the EME system was compromised. In a separate trial, 1-octanoic acid was replaced by 1-decanoic acid to enhance the hydrophobicity of the liquid membrane. However, even with this modification, the extraction current remained high, and the EME system continued to exhibit instability. This observation aligns with findings in previous literature regarding aromatic solvents comparable to thymol in terms of polarity, where poor extraction efficiency

**Table 1**
Physicochemical properties and structure of eutectic solvent constituents.

| | Log P | HBD | HBA | Aromat | $pK_a$ | Structure |
|---|---|---|---|---|---|---|
| Thymol | 3.43 | 1 | 1 | 1 | 10.59 |  |
| Menthol | 2.66 | 1 | 1 | 0 | – |  |
| Camphor | 2.55 | 0 | 1 | 0 | – |  |
| Octanoic acid | 2.7 | 1 | 2 | 0 | 5.19 |  |
| Decanoic acid | 3.59 | 1 | 2 | 0 | 4.95 |  |

was attributed to fouling of the aromatic liquid membranes [31]. Given these challenges and based on insights from prior studies, further investigation of liquid membranes based on thymol was not pursued in the present study.

### 3.3. Extraction with menthol as hydrogen bond acceptor

In a subsequent series of experiments, we conducted extractions using a deep eutectic solvent synthesized from menthol and 1-octanoic acid. In contrast to thymol, the menthol-based liquid membrane could not extract peptides from buffer solutions, as depicted in Table S2 (supporting information). Notably, all peptides, except endomorphin-1 remained within the sample after extraction.

The Kamlet-Taft parameters for menthol-octanoic acid were measured (according to Ref. [30] to $\alpha = 0.6$, $\beta = 0.43$, and $\pi^* = 0.39$. The relatively high α-value indicated that the membrane solvent provided sufficient hydrogen bond acidity. However, cation-π interactions were absent in the menthol-octanoic acid solvent owing to the absence of aromatic π-electron systems. Evidently, the hydrogen bond basicity (β) of the membrane solvent was insufficient, and the peptides remained in the sample during extraction.

In a final attempt to enhance the extraction efficiency, ionic interactions were introduced into the system by addition of 2.0% di (2-ethylhexyl) phosphate (DEHP) to the menthol-octanoic acid liquid membrane by weight. However, even with DEHP, the extraction efficiency remained notably constrained. Consequently, further experiments utilizing menthol-based DES were not pursued.

### 3.4. Extraction with camphor as hydrogen bond acceptor

In the final series of initial experiments, extractions were conducted employing a deep eutectic solvent prepared from camphor and 1-decanoic acid. Kamlet-Taft parameters for this composition were measured to $\alpha = 0.65$, $\beta = 0.62$, and $\pi^* = 0.55$. The extraction recoveries obtained using this particular liquid membrane are summarized in Table 2. Notably, six peptides were extracted using these conditions.

To enhance extraction efficiency, 2% DEHP was introduced into the camphor-decanoic acid liquid membrane. This modification led to a significant improvement in the extraction efficiency. As depicted in Table 2, all 13 model peptides were extracted from the buffer solution, and seven of them had recoveries exceeding 85%. Utilizing this liquid membrane for plasma samples, nine peptides were extracted with recovery above 40%, either at pH 3.0 or 6.0 (Table 2). Four peptides were extracted with recoveries in the range of 10–40%.

**Table 2**
Extraction recoveries with liquid membranes based on camphor and 1-decanoic acid.

| Peptides | Camphor-decanoic acid (1:1 w/w) Buffer 30 V 30 min | Camphor-decanoic acid (1:1 w/w) 2%DEHP Buffer 30 V 30 min | Camphor-decanoic acid (1:1 w/w) 2%DEHP Plasma-pH 3.0 15 V 45 min | Camphor-decanoic acid (1:1 w/w) 2%DEHP Plasma-pH 6.0 15 V 45 min |
|---|---|---|---|---|
| Vasopressin | n.d. | 25% | n.d. | 42% |
| Diprotin B | 10% | 40% | 10% | n.d. |
| Bradykinin | n.d. | 100% | 15% | 73% |
| Diprotin A | 20% | 60% | 41% | n.d. |
| Angiotensin III | n.d. | 100% | 52% | 58% |
| Angiotensin II | n.d. | 100% | 18% | 16% |
| Met-enkephalin | 15% | 60% | 43% | n.d. |
| Angiotensin IV | 15% | 100% | 72% | 55% |
| Oxytocin | n.d. | 10% | n.d. | 17% |
| Neurotensin | n.d. | 85% | 8% | 25% |
| Angiotensin I | n.d. | 85% | 45% | 70% |
| Leu-enkephalin | 50% | 95% | 44% | 11% |
| Endomorphin-1 | 100% | 85% | 86% | 45% |

### 3.5. Extraction current data

The recorded current profiles not only served as a valuable indicator of system stability, as discussed earlier, but also yielded insights into the overall ion flux across the liquid membrane. As depicted in Table 3, the current observed in the thymol-based systems from buffer solutions was notably lower in comparison to those based on camphor. This disparity can be attributed to the lower polarity of thymol as opposed to camphor, resulting in a reduced flux of buffer ions within the thymol-based systems.

However, from plasma samples, the thymol-based systems were highly influenced by the plasma matrix. The case of thymol-decanoic acid-DEHP is particularly noteworthy. This liquid membrane is capable of hydrogen bond, cation-π, and ionic interactions. For this reason, the overall ion flux from plasma and across the liquid membrane was high, and the extraction system was not sufficiently stable.

### 3.6. Effect of sample pH

Based on the results reported above, camphor-decanoic acid-DEHP was selected for further optimization. In a first set of optimization experiments, we investigated the relationship between extraction recoveries and the pH of the sample solution. The findings are summarized in Fig. 2, focusing on four representative model peptides. The trend observed for vasopressin (GRAVY index = −0.78, approximate value, see Table S1) was indicative of the behavior of bradykinin (GRAVY index = −1.0). Both peptides were relatively polar, and are nine amino acids long. The optimal sample pH range was identified as pH 6.0–7.0. In this pH range, both peptides were net positively charged, corresponding to +2, and the recoveries exceeded 40%. Below pH 6.0, the recoveries declined rapidly with decreasing pH, leading to a noticeable discrimination.

This phenomenon was attributed to the formation of a negative boundary layer at the interface between the liquid membrane and the sample (Fig. 3), induced by the deprotonation of 1-decanoic acid ($pK_a$ = 4.8). At pH 6.0–7.0, the presence of a negative boundary layer facilitated ionic interactions with vasopressin and bradykinin, and increased their mass transfer into the liquid membrane. At pH-values below 6.0, the local deprotonation of 1-decanoic acid was suppressed, the negative boundary layer disappeared, and mass transfer was reduced. Oxytocin (GRAVY index = 0.33, approximate value, see Table S1) showed similar pattern, but this peptide was extracted with lower recovery (≈20%). Oxytocin is nine amino acids long and contains only one basic functionality (at the N-terminal). Due to low charge-to-size ratio, solvation based on ionic interactions was limited and this limited the extraction efficiency.

The behavior of leu-enkephalin (GRAVY index = 0.90) in Fig. 2 was representative of met-enkephalin (GRAVY index = 0.52), diprotin B (GRAVY index = 2.1), diprotin A (GRAVY index = 2.47), and endomorphin-1 (GRAVY index = −0.25, approximate value, see Table S1). These are less polar peptides of three to five amino acids. At

**Table 3**
Average extraction current for each EME extraction cell.

| | Current (μA/V) | |
|---|---|---|
| | Buffer solution | Plasma sample |
| Camphor-decanoic acid | 0.06 | 0.06 |
| Camphor-decanoic acid-DEHP | 0.2 | 0.3 |
| Thymol-decanoic acid | 0.02 | 0.7 |
| Thymol-decanoic acid-DEHP | 0.08 | 33 |

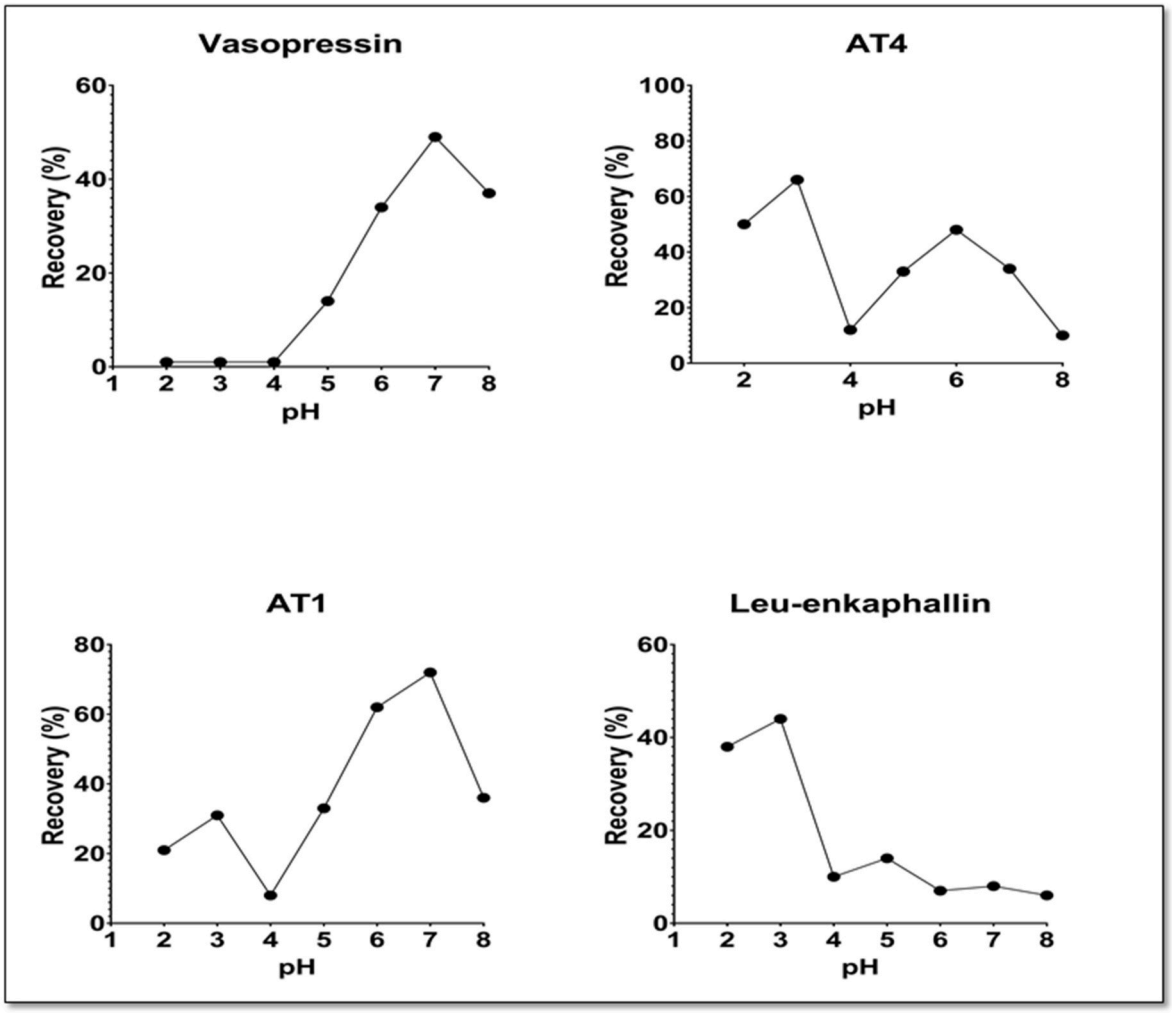



**Fig. 2.** Extraction recovery as a function of pH in the plasma sample solution (acceptor pH 2, voltage and extraction time: 30 V and 30 min).

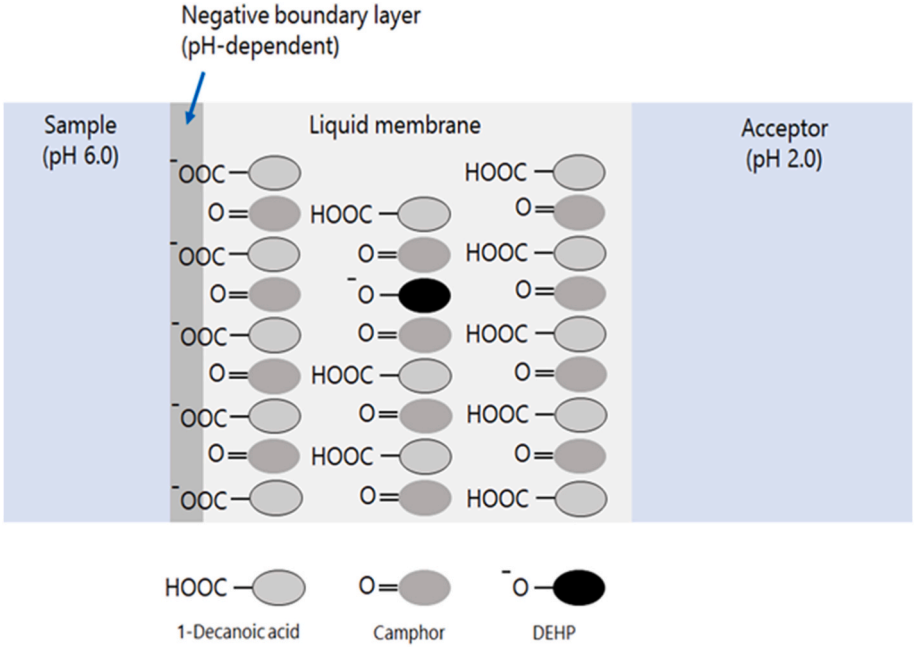



**Fig. 3.** Negative boundary layer at liquid membrane-sample interface.

pH 6.0, all these peptides were under isoelectric conditions (except endomorphin-1), and recoveries were therefore less than 10%. Optimal conditions were obtained at pH 3.0. Both leu-enkephalin (net charge +1), met-enkephalin (net charge +1), diprotin A (net charge +0.2), and endomorphin-1 (net charge +1) were extracted with recoveries exceeding 40%. The recovery for diprotin B (net charge +0.2) was below 40%, and this was attributed to slightly higher polarity than diprotin A.

The behavior of angiotensin IV (Fig. 2) was representative of angiotensin II and III. Extraction recoveries demonstrated an optimum range at pH 6.0–7.0, where the peptide net charges were in the range of $+0.5 < z < +1.5$. As the pH of the sample solution decreased, the recoveries decreased significantly because the negative boundary layer disappeared. Intriguingly, below pH 4.0, recoveries increased again and were optimum at pH 3.0. At this low sample pH, the net charge of the peptides was in the range $+1.2 < z < 2.2$. Angiotensin III (GRAVY index = 0.13) and angiotensin IV (GRAVY index = 0.90) were extracted with recoveries exceeding 40%. Due to polarity, the recovery of angiotensin II (GRAVY index = −0.32) was less than 40%.

The optimum sample pH for angiotensin I (Fig. 2) and neurotensin was found at pH 7.0. Interestingly, at this pH, both peptides are very close to isoelectric conditions ($z<+0.2$), and still angiotensin I (GRAVY index = −0.20) was extracted with a recovery exceeding 40%. Most probably, sample pH is somewhat lower than 7.0 in the sample-liquid membrane boundary layer, and this in combination with the negative boundary layer induced by 1-decanoic acid favored mass transfer.

### 3.7. Optimization

In subsequent experiments, we investigated the influence of pH in the acceptor phase. The results for four selected peptides are presented in Fig. 4. Recoveries were highest at pH 2.0–3.0, and decreased as the pH increased from 3.0 to 4.0. This pattern is typical with DEHP [24,27]. DEHP has a $pK_a$-value of 1.94, and for efficient release of peptides at the liquid membrane/acceptor interface, a low pH was favorable for suppressing ionic interactions. As pH was increased above 4.0, a negative boundary layer was established at the acceptor-liquid membrane interface, and this was balanced by protons from the acceptor. Thus, the boundary layer was acidified, which increased the extraction efficiency again.

We then conducted extractions under various extraction potentials within the voltage range of 0–30 V. Generally, the extraction recoveries exhibited an upward trend with increasing voltage, reaching an optimum value at approximately 15 V. Beyond this threshold, the system displayed reduced stability, prompting us to designate 15 V as the optimal extraction potential (Fig. S1, supporting information).

Finally, we optimized the extraction time, and as expected, the extraction recoveries exhibited a gradual increase with extension of time, reaching their maximum at 45 min. Beyond this time frame, the system reached a state of equilibrium, and the recoveries no longer increased (Fig. S2, supporting information). The kinetics of the extraction process for peptides were relatively slow compared to the EME of small molecule pharmaceuticals [2].

### 3.8. Evaluation

The optimized EME procedure was then evaluated for quantification of the model peptides in human plasma. The method evaluation included limits of detection (LOD), limits of quantification (LOQ), repeatability and reproducibility (RSD), and linear dynamic range (LDR). Results are summarized in Table 4. Based on three replicate experiments (concentration 100 ng/mL), all RSDs were between 3% and 16% for intra-day experiments, and between 4% and 16% for inter-day experiments. When using one of the analytes (AT III) as an internal standard, the RSDs were significantly reduced (by a factor of three). The limits of detection (S/N = 3) and quantification (S/N = 10) were in the ranges of 0.01–12.5 and 0.03–42.0 ng/mL, respectively.

In this study, we conducted a comparative analysis of the performance of the current liquid membrane with another liquid membrane

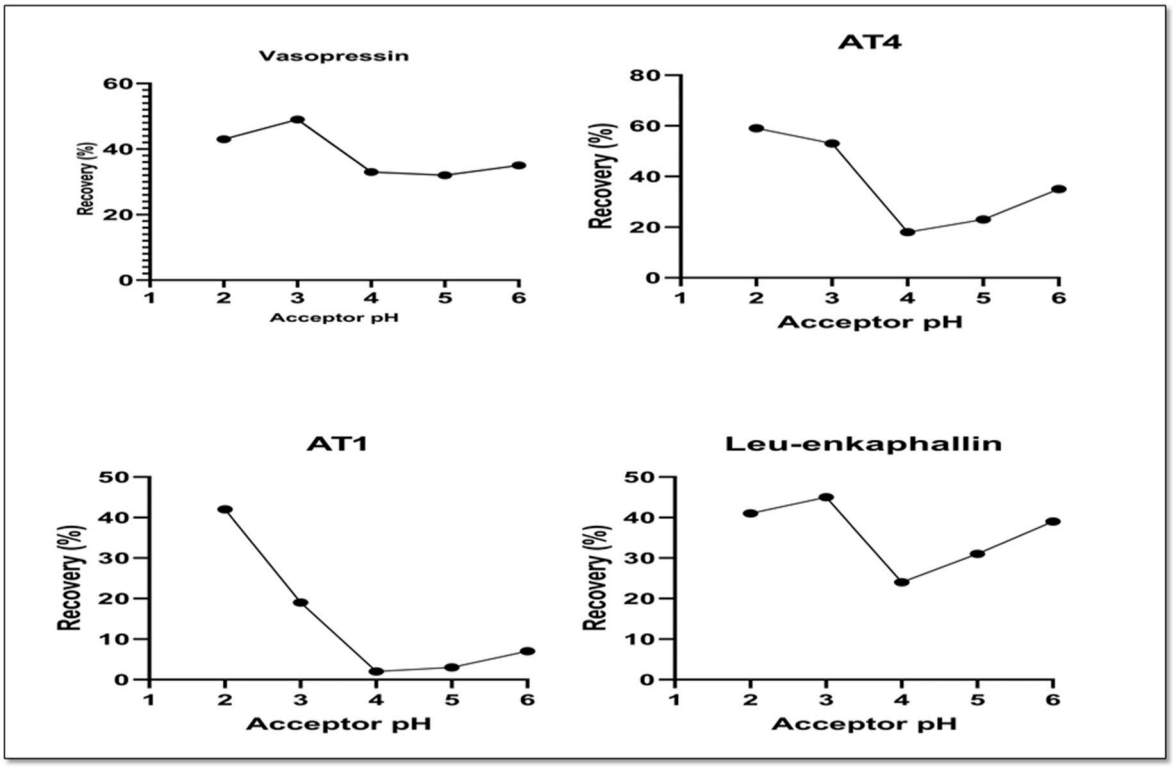



**Fig. 4.** Extraction recovery from plasma as a function of pH in the acceptor (donor pH 6 for vasopressin and 3 for Angiotensin I, IV and Leu-enkaphallin, voltage and extraction time: 30 V and 30 min).

**Table 4**
The analytical characteristics of the method from human plasma.

| | Linearity range[a] | Quantification limit[a] | Detection limit[a] | Recovery[b] | | Repeatability[c] | | Reproducibility[c] | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | pH3 | pH6 | pH3 | pH6 | pH3 | pH6 |
| Vasopressin | 50–5000 | 25.0 | 7.50 | n.d. | 25 | – | 16 | 6 | 16 |
| Diprotin B | 50–2000 | 42.0 | 12.5 | 20 | n.d. | 5 | – | 16 | 15 |
| Bradykinin | 50–5000 | 33.0 | 10.0 | 20 | 70 | 4 | 5 | 10 | 10 |
| Diprotin A | 0.5–2000 | 0.35 | 0.10 | 35 | n.d. | 4 | – | 14 | 11 |
| Angiotensin III | 0.1–1000 | 0.08 | 0.03 | 44 | 55 | 9 | 7 | 16 | 7 |
| Angiotensin II | 0.2–2000 | 0.03 | 0.01 | 27 | 20 | 5 | 12 | 13 | 12 |
| Met-enkephalin | 0.5–2000 | 0.40 | 0.12 | 27 | n.d. | 5 | – | 14 | 16 |
| Angiotensin IV | 50–10000 | 22.0 | 7.00 | 50 | 20 | 4 | 4 | 12 | 4 |
| Oxytocin | 1.0–10000 | 0.80 | 0.25 | n.d. | 10 | – | 3 | 14 | 7 |
| Neurotensin | 5.0–5000 | 2.50 | 0.72 | 10 | 30 | 9 | 6 | 14 | 9 |
| Angiotensin I | 0.2–5000 | 0.10 | 0.03 | 53 | 45 | 6 | 5 | 16 | 13 |
| Leu-enkephalin | 0.5–5000 | 0.30 | 0.10 | 36 | 20 | 4 | 7 | 9 | 7 |
| Endomorphin-1 | 1.0–10000 | 0.20 | 0.06 | 75 | 40 | 13 | 12 | 11 | 9 |

[a] All concentration are ng/mL.
[b] Concentration level = 100 ng/mL.
[c] RSD %, n = 3 (These data are without internal standard correction).

based on a deep eutectic solvent (coumarin and thymol (1:2) + 2% (v/v) di (2-ethylhexyl) phosphate (DEHP)), previously published by our research group for extraction the same model peptides in 96-well format [29].

Both methods utilized an environmentally friendly liquid membrane composed of a deep eutectic solvent, aligning with green principles in sample preparation. An essential advantage of the present liquid membrane was its ability to offer a stable extraction. The commercially available EME system emerged as a robust and efficient method for extraction of peptides from plasma solutions. There were no significant changes observed in terms of equilibrium time and recoveries.

Furthermore, we compared this commercialized EME system with gel membrane EME system for the extraction of different peptides from aqueous and plasma samples. While gel EME represented a greener extraction system compared to the current method, the recoveries achieved with the current method were notably higher. Additionally, the sample volume required was reduced, making this method particularly advantageous for applications that necessitate minimal sample consumption [20,22].

### 3.9. LC-MS/MS and matrix effects

We first checked for co-extraction of phospholipids into the acceptor, according to a previously described procedure [32]. We compared EME extracts from plasma obtained with camphor-decanoic acid-DEHP and NPOE. Additionally, we compared the EME extracts with protein precipitated samples. Protein precipitation was performed using acetonitrile as the precipitant. As depicted in Fig. 5, no phospholipids were detected in any of the acceptor solutions, underscoring the successful provision of phospholipid-free extracts by the EME system. Next,

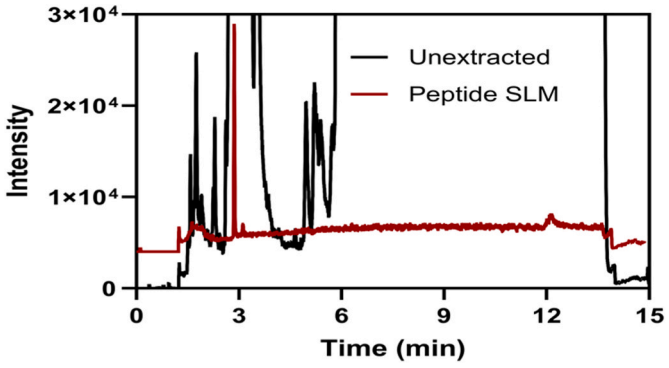



**Fig. 5.** LC-MS/MS analysis of phospholipids in the acceptor phase.

leakage of camphor and decanoic acid into the acceptor. The content was quantified using an LC-UV method, and calibrated against dilutions of the SLM itself. After extraction, 11.12 ± 0.04% of the camphor in the SLM leaked into the acceptor, while only 0.22 ± 0.02% of decanoic acid leaked (n = 3). This was in agreement with the intrinsic solubility of each component.

Finally, we checked for matrix effects caused by matrix components or residues of camphor or decanoic acid, and the relevant data is presented in Table S3. The Matrix Effect (ME) values for all extracted peptides, both at pH 3.0 and 6.0, consistently fell within the range of 71–109%. Notably, this EME system demonstrated exceptional efficiency in the context of target analytes from a complex matrix, as the extracted peptides were reliably detected without encountering ion suppression. These findings underscore the feasibility of conducting quantification without any interference from other compounds present in the acceptor phase.

## 4. Conclusions

In the present work, we have developed a new type of liquid membrane for electromembrane extraction of peptides, based on a deep eutectic solvent of camphor and 1-decanoic acid, and containing 2% di (2-ethylhexyl) phosphate (DEHP) as ionic carrier. In contrast to EME systems for peptides published previously, this new liquid membrane provided a stable extraction system from human plasma, with limited leakage of liquid membrane constituents into the acceptor. The system was operated at current levels well below 50 μA, to avoid pH changes and increased pressure due to excessive electrolysis. Recoveries exceeding 40% was obtained for nine of 13 model peptides included in our training set of analytes.

Model peptides with more than one base functionality (including the N-terminal) and with GRAVY index > −1 were all extracted from plasma with recoveries exceeding 40%. Similar performance was also obtained for peptides with a single base functionality only (in the N-terminal) and with GRAVY index > 0.5. For the tri- and tetra-peptides studied (one base functionality), those with log P > −0.25 were extracted with recoveries exceeding 40%.

EME is an inexpensive and green sample preparation technique. The present work has provided important insight into the fundamentals of EME. The camphor-decanoic acid-DEHP membrane provided an appropriate balance between hydrogen bond acidity and basicity, to assist the solvation of peptides with both negative and positive charges. In addition, solvation was facilitated by ionic interactions with DEHP. When camphor was replaced by the aromatic analogue thymol, the flux of matrix ions from plasma increased dramatically due to π-type of interactions, and the extraction system became unstable. Work is in

progress with the camphor-decanoic-DEHP liquid membrane, to investigate the potential for increasing selectivity. We foresee that EME can be developed into a highly valuable tool for extraction of peptides in biological fluids, including signature peptides originating from proteins subjected to enzymatic digestion.

## CRediT authorship contribution statement

**Samira Dowlatshah:** Writing – review & editing, Writing – original draft, Validation, Formal analysis, Conceptualization. **Torstein Kige Rye:** Writing – review & editing, Formal analysis. **Frederik André Hansen:** Writing – review & editing, Writing – original draft, Supervision, Formal analysis, Data curation, Conceptualization. **Trine Grønhaug Halvorsen:** Writing – review & editing, Supervision, Conceptualization. **Stig Pedersen-Bjergaard:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Conceptualization.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

Data will be made available on request.

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## Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.aca.2024.342360.

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